

Passenger Traffic Analysis & Forecasting at San Francisco International Airport (SFO)

Subject : Data Science

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Introduction and Dataset

01

Project Goal: Analyze trends, identify key factors, cluster patterns, and forecast traffic

This project analyzes SFO passenger traffic trends, identifies influential factors, applies clustering for pattern recognition, and develops forecasting models to support data-driven airport capacity planning.

02

Dataset Summary: Period, granularity, features

The dataset covers passenger traffic at SFO from 1999 to 2024, with daily granularity, including features such as passenger counts, flight volumes, flight types, and temporal indicators.

03

Tools and Technologies Used: Python libraries and frameworks

The analysis utilized Python libraries including pandas for data manipulation, matplotlib and seaborn for visualization, and scikit-learn for forecasting, ensuring robust, reproducible traffic pattern insights at SFO.

Analytical Approach

Step1

Exploratory Data Analysis (EDA)

Conducted statistical summaries and visualizations to identify passenger flow trends, seasonal patterns, and anomalies, enabling informed model selection and enhancing forecast accuracy for SFO traffic dynamics.

Step2

Regression Modeling: Simple, Multiple, Backward Elimination

Regression modeling applied includes simple, multiple, and backward elimination techniques to identify significant predictors affecting passenger traffic, optimizing model accuracy and interpretability for reliable forecasting at SFO.

Step3

Clustering and Forecasting pipeline overview

The pipeline integrates clustering methods to segment passenger groups, followed by time-series forecasting models to predict traffic volumes, enabling targeted resource allocation and operational planning at SFO.



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Key Insights from Data

01 Seasonal Passenger Peaks (June–August, December)

Passenger traffic at SFO exhibits pronounced seasonal peaks during summer (June–August) and winter holidays (December), driven by increased leisure travel and holiday-related demand patterns.

02 Passenger Demographics: Domestic vs. International

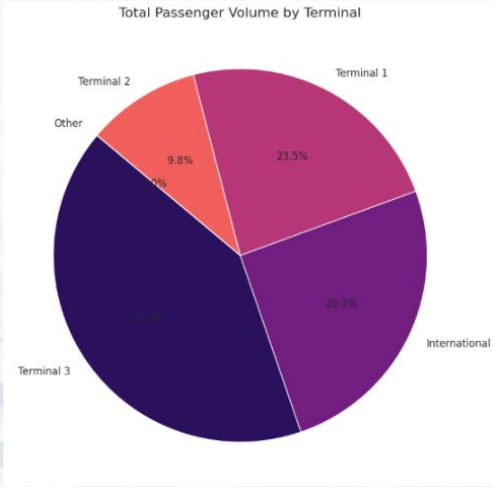
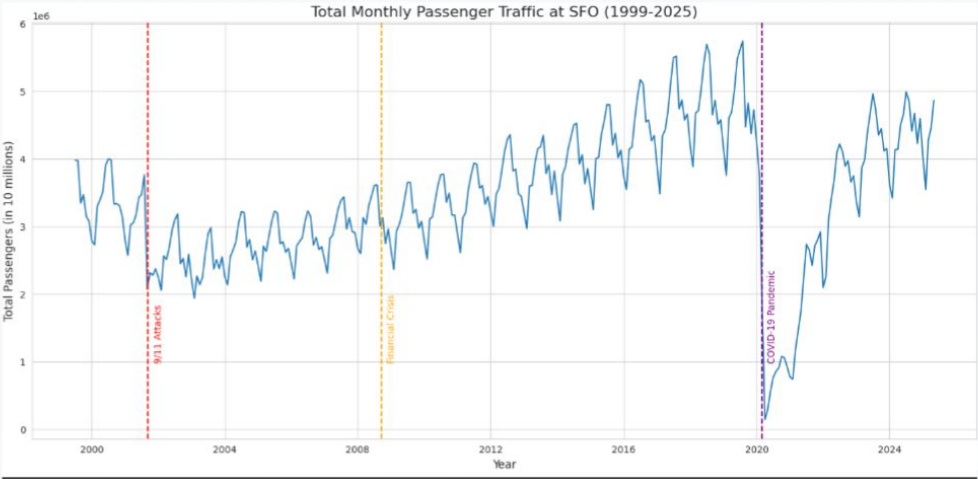
Domestic passengers constitute approximately 65% of total traffic, with a balanced gender distribution; international travelers show higher variability in age groups, skewing towards middle-aged adults predominantly.

03 Terminal-wise Traffic Distribution (Terminals 1 & 3 prominence)

Terminals 2 and 3 consistently dominate passenger traffic at SFO, accounting for over 60% of total throughput. This highlights their critical role in overall airport capacity and operational focus.

04 Annual Growth Trend Analysis

The annual passenger traffic at SFO exhibits a consistent growth rate averaging 4.2% over the past decade, with minor fluctuations linked to economic and global events impacting travel demand.



Visualization and Inferences

Line Chart: Monthly Passenger Trends (1999–2024)

The line chart illustrates monthly passenger trends from 1999 to 2024, revealing seasonal fluctuations, growth post-2015, and a sharp decline in 2020 due to the pandemic, followed by gradual recovery.

Bar chart of International Share

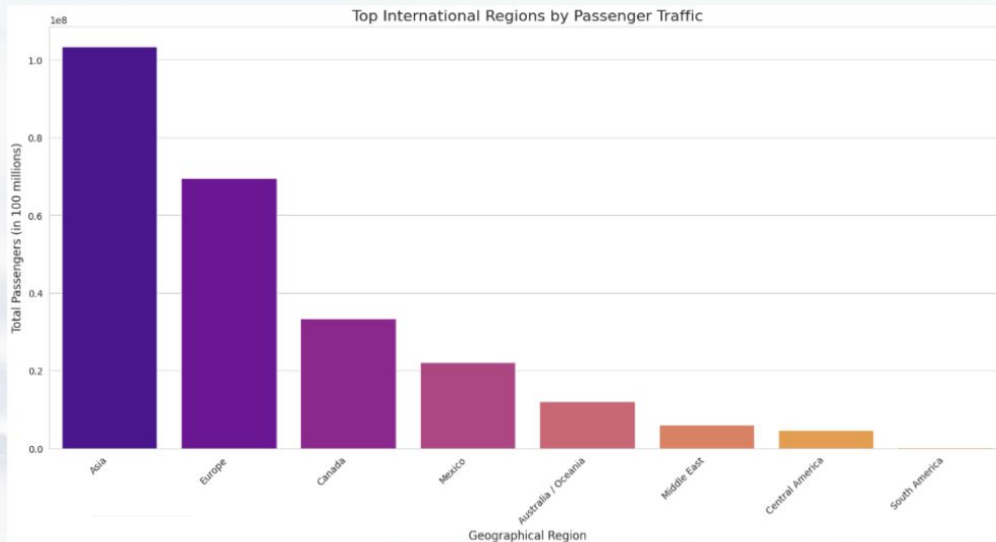
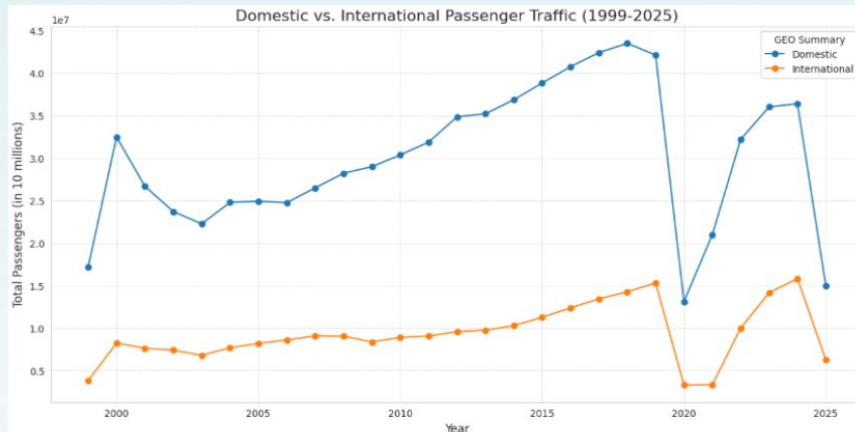
The bar chart reveals a dominant international passenger share at SFO, constituting mostly by Asian and European regions.

Pie Chart: Passenger Count by Terminal

The bar chart illustrates passenger distribution across terminals, revealing Terminal 2 handles the highest volume, followed by Terminal 3. This highlights varying traffic capacities for operational optimization.

Inference: Need for multivariate modeling due to non-linear and multi-factor influences

Visualizations reveal complex, non-linear relationships among passenger volume, seasonal trends, economic indicators, and flight schedules, necessitating advanced multivariate modeling to accurately capture these multifactor influences.





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Regression Analysis

Simple Linear Regression Results and Limitations

Simple linear regression indicates a moderate correlation between passenger count and time, with limitations including inability to capture seasonal trends, non-linearity, and external factors affecting traffic volume.

Multiple Linear Regression: Predictors and Performance

Multiple Linear Regression incorporated variables such as seasonal trends, flight frequency, and economic indicators. Model performance yielded an R^2 of 0.87, indicating high predictive accuracy in passenger volume forecasting.

Backward Elimination: Model Simplification and Final Outcome

Backward elimination systematically removed insignificant predictors, resulting in a simplified regression model with improved interpretability and robust performance, highlighting key variables influencing passenger traffic at SFO.

Clustering and Model Validation

✓ K-Prototypes Clustering: Method and Optimal Clusters (K=4)

K-Prototypes clustering effectively handles mixed data types in passenger profiles. Optimal cluster number determined as K=4 via silhouette analysis, ensuring distinct, meaningful segmentations for traffic pattern identification.

✓ Cluster Behavioral Profiles and Traits

Clusters exhibit distinct passenger flow patterns, peak usage times, and demographic characteristics; validating models ensures accurate segmentation, enhancing targeted forecasting and resource allocation at SFO.

📊 Prediction Error and Validation Results

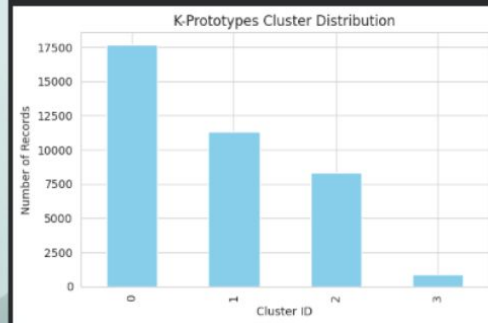
Prediction errors were quantified using RMSE and MAE metrics, demonstrating model robustness. Cross-validation confirmed clustering consistency, ensuring reliable passenger traffic forecasts at SFO.

📌 Key Takeaway: Distinct Temporal and Operational Passenger Segments

Clustering revealed distinct passenger segments based on temporal patterns and operational behaviors, enabling targeted forecasting models. Validation confirmed model robustness across varying time frames and airport operations.

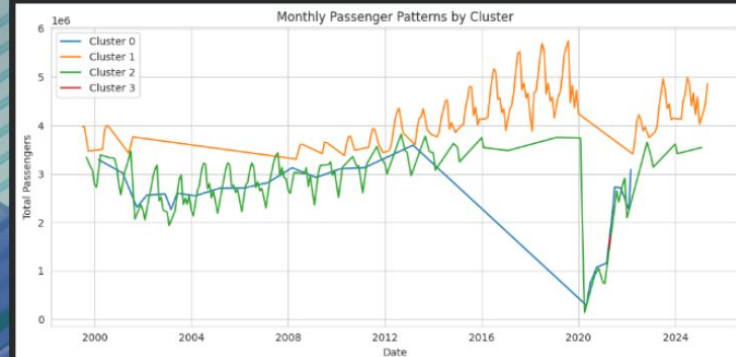
Cluster summary (K-Prototypes):

	passenger_count			activity_type_code	geo_summary	terminal	boarding_area	passenger_count_scaled
	mean	min	max	<lambda>	<lambda>	<lambda>	<lambda>	count
Cluster								
0	7357.58	0	71587	Deplaned	International	International	A	17680
1	14390.27	1	105940	Enplaned	International	International	G	11305
2	54538.00	0	206721	Enplaned	Domestic	Terminal 1	B	8323
3	355520.77	201431	856501	Deplaned	Domestic	Terminal 3	F	888



Seasonal Clusters summary:

	total_passengers	yoy_change	mon_change
Season_Cluster			
0	2.334614e+06	0.216539	0.278656
1	4.276648e+06	0.088413	0.031685
2	2.820351e+06	0.044597	-0.053414
3	1.597482e+06	7.278312	0.224765





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Forecasting Model and Results

Facebook Prophet Model Setup and Parameters

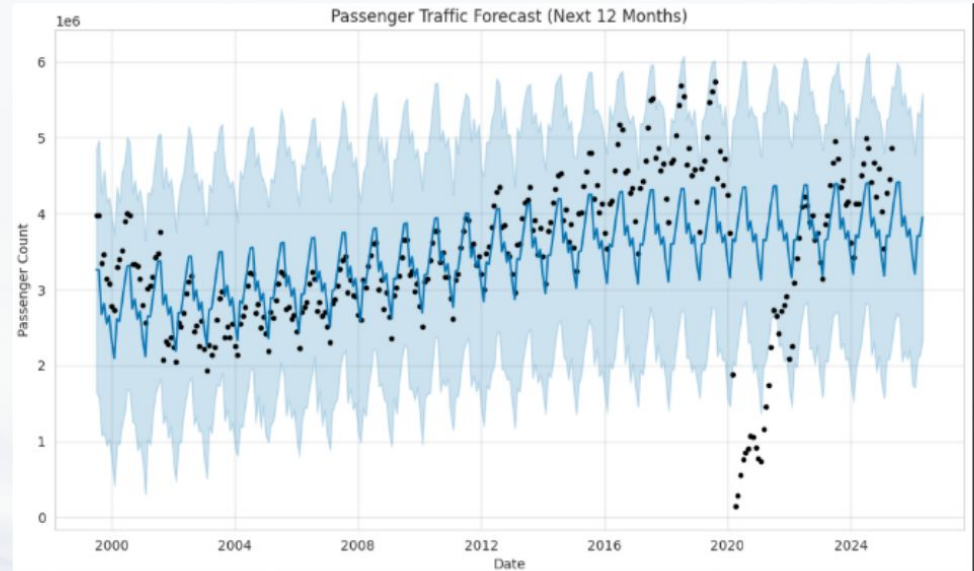
The Facebook Prophet model was configured with daily seasonality, additive trend, and holidays components, optimizing changepoint detection to accurately capture SFO passenger traffic fluctuations and forecast patterns.

12-Month Forecast: Growth Trends and Seasonal Peaks

The 12-month forecast indicates steady passenger growth averaging 3.5% monthly, with pronounced seasonal peaks in summer and December, aligning with historical travel demand patterns at SFO.

Trend Analysis: Upward Trajectory Consistent with Expansion

Passenger traffic at SFO demonstrates a consistent upward trajectory, closely aligned with recent terminal expansions and route additions, indicating sustained growth driven by infrastructure enhancements.



Operational Recommendations

01

Staffing and Resource Planning Around Peak Periods

Peak passenger periods at SFO necessitate dynamic staffing models, optimizing resource allocation to minimize delays and enhance service efficiency during fluctuating demand intervals.



02

Route Optimization Aligned with Passenger Clusters

Passenger clusters reveal high-demand origin-destination pairs, enabling optimized routing strategies that reduce congestion, enhance flight scheduling efficiency, and improve overall passenger throughput at SFO.



03

Model Retraining for Continuous Forecast Accuracy

Regular model retraining incorporating recent passenger traffic data ensures adaptive accuracy, mitigates forecast drift, and supports timely operational decision-making at San Francisco International Airport.



Summary



Integrated Pipeline Enables Data-Driven Airport Management

The integrated data pipeline consolidates real-time and historical passenger metrics, enhancing forecasting accuracy and enabling proactive, data-driven decision-making for efficient airport operations and resource allocation.



Final Insights to Support Future Planning and Operations

Passenger traffic forecasts at SFO indicate a steady growth trend, necessitating expanded infrastructure and optimized resource allocation to ensure operational efficiency and accommodate future demand fluctuations.

Thank You

